HW8 杜乘旋 12209095

1. (a) Pareto optimum or the set of pareto optima is a Resource Allocation that is no such deviation from Pareto optimum that the deviated Resource Allocation makes at least one agent better off without hurting someone else in the economy. It's not unique. Perfect risk sharing is not achieved, as both agents have their individual risk either at stage $\theta=1$ or $\theta=2$

If state-2 endowments were 5 for agent 1 and 3 for agent 2. The answer remains the same

(b) for agent 1: $\max_{C_0, C_1(\theta)} 0.25C_0 + \frac{1}{4}\ln(C_1(1)) + \frac{1}{4}\ln(C_1(2))$

s.t. $2+2q_1+4q_2 \geq C_0 + q_1 C_1(1) + q_2 C_1(2)$

$$L_1 = 0.25C_0 + \frac{1}{4}\ln(C_1(1)) + \frac{1}{4}\ln(C_1(2)) + \lambda_1(2+2q_1+4q_2 - C_0 - q_1C_1(1) - q_2C_1(2))$$

F.O.C. $\begin{cases} \lambda_1 = 0.25 \\ \frac{1}{4C_1(1)} - \lambda_1 q_1 = 0 \\ \frac{1}{4C_1(2)} - \lambda_1 q_2 = 0 \end{cases} \Rightarrow \begin{cases} C_0 = 2q_1 + 4q_2 \\ C_1(1) = \frac{1}{q_1} \\ C_1(2) = \frac{1}{q_2} \end{cases}$

$$L_2 = C_0 + \frac{1}{2}\ln(C_1(1)) + \frac{1}{2}\ln(C_1(2)) + \lambda_2(4+4q_1+2q_2 - C_0 - q_1C_1(1) - q_2C_2(2))$$

F.O.C $\begin{cases} \lambda_2 = 1 \\ \frac{1}{2C_1(1)} - \lambda_2 q_1 = 0 \\ \frac{1}{2C_1(2)} - \lambda_2 q_2 = 0 \end{cases} \Rightarrow \begin{cases} C_0 = 3+4q_1+2q_2 \\ C_1(1) = \frac{1}{2q_1} \\ C_2(2) = \frac{1}{2q_2} \end{cases}$

$\therefore \begin{cases} \frac{3}{2q_1} = 6 \\ \frac{3}{2q_2} = 6 \end{cases} \quad \therefore \begin{cases} q_1 = \frac{1}{4} \\ q_2 = \frac{1}{4} \end{cases}$

		t=0	t=1 θ=1	t=1 θ=2
CERA:	Agent 1	1.5	4	4
	Agent 2	4.5	2	

It's pareto optimal; the determinants of q_i depends on the probability of future state the utility function of investors and aggregate wealth of the future state.

(c) introduce 2 with payoff (1, 0)

for agent 1: change of u() after trading is:

$1.5\times 0.25 + 0.5\times 0.5\cdot \ln 4 - (2\times 0.25 + 0.5\times 0.5\ln 2) \approx 0.04$

for agent 2: change of u() after trading: $0.5 - \frac{1}{2}\ln 4 + \frac{1}{2}\ln 2 \approx 0.15$ $0.04 < 0.15$

For Agent 2 this introduction is more valuable

The outcome is Pareto optimal for R.A is the same as CERA.

2. (a) This two "complex" assets must be the portfolio of the contingent claim

$$\begin{cases} 2q_1+4q_2=2 \\ q_1+3q_2=1.25 \end{cases} \Rightarrow \begin{cases} q_1=0.5 \\ q_2=0.25 \end{cases}$$

(b) $P_b^0 = q_1+q_2 = 0.75$

(c) $P_s^0 = 2q_1+3q_2 = 1.75$

3. ①

$$\begin{pmatrix} 1 & 2 & 9 \\ 3 & 7 & 8 \\ 7 & 16 & 25 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 9 \\ 0 & 1 & -19 \\ 0 & 2 & -38 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 9 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{pmatrix}$$

linearly dependent : not complete

$$\text{states} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \begin{pmatrix} A & B & C \\ 3 & 7 & 0 \\ 1 & 2 & 0 \\ 7 & 16 & 16-k \end{pmatrix}$$ is complete

A-D securities $\begin{cases} \text{states-1: } B-2A-\frac{2}{16-k}C\begin{pmatrix}1\\0\\0\end{pmatrix} \\ \text{states-2: } 7A-3B-\frac{1}{16-k}C\begin{pmatrix}0\\1\\0\end{pmatrix} \\ \text{states-3: } \frac{1}{16-k}C\begin{pmatrix}0\\0\\1\end{pmatrix} \end{cases}$

where C is a call on B

[Suppose 7ck of B<16)

reproduce: $\begin{cases} A=3S_1+1S_2+7S_3 \\ B=7S_1+2S_2+16S_3 \\ C=(16-k)S_3 \end{cases}$

②

$$\begin{pmatrix} 1 & 5 & 12 \\ 3 & 6 & 14 \\ 12 & 7 & 16 \\ 15 & 8 & 20 \end{pmatrix} \to \begin{pmatrix} 1 & 5 & 12 \\ 0 & -9 & -22 \\ 0 & -53 & -128 \\ 0 & -67 & -160 \end{pmatrix} \to \begin{pmatrix} 1 & 5 & 12 \\ 0 & 1 & \frac{22}{9} \\ 0 & 0 & \frac{14}{9} \\ 0 & 0 & \frac{34}{9} \end{pmatrix}$$

only 3 pivots : not complete

$$\text{states} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{pmatrix} A & B & C & D \\ 1 & 5 & 12 & 0 \\ 3 & 6 & 14 & 0 \\ 12 & 7 & 16 & 0 \\ 15 & 8 & 20 & 20+k \end{pmatrix}$$ is complete

A-D securities $\begin{cases} \text{state-1: } \frac{1}{7}A-\frac{60}{7}B+\frac{57}{14}C-\frac{45}{7(20-k)}D\begin{pmatrix}1\\0\\0\\0\end{pmatrix} \\ \text{state-2: } -\frac{2}{7}A+\frac{64}{7}B-\frac{53}{14}C+\frac{48}{7(20-k)}D\begin{pmatrix}0\\1\\0\\0\end{pmatrix} \\ \text{state-3: } \frac{1}{7}A-\frac{11}{7}B+\frac{9}{14}C-\frac{12}{7(20-k)}D\begin{pmatrix}0\\0\\1\\0\end{pmatrix} \\ \text{state-4: } \frac{1}{20-k}D\begin{pmatrix}0\\0\\0\\1\end{pmatrix} \end{cases}$

Where D is a call on C

(suppae 16ck of C<20)

reproduce: $\begin{cases} A=1S_1+3S_2+12S_3+15S_4 \\ B=5S_1+6S_2+7S_3+8S_4 \\ C=12S_1+14S_2+16S_3+20S_4 \\ D=\frac{1}{20-k}S_4 \end{cases}$